

the option that has the same current requirement but is costlier from further analysis.

In our next stage, we investigate the design options for driving the relay. Most microcontrollers can supply about 15mA continuously. Solid-state relays can be driven directly from the microcontroller, relays that need about 100mA can be driven by a transistor, and for higher current requirements we can use a MOSFET switching transistor.

We add the input cost with the cost of the device and list it with the option. For example, we need 7.5mA current for a solid-state relay. We got the cost of the relay as ₹2,783. So, the total cost till Stage 2 for this option works out to be ₹2,783. For relays that require 75mA current, we need a transistor, resistor, and a flyback diode. The total cost of these is ₹20, and the net cost till Stage 2 comes out to be ₹170.

We evaluate the other options in a similar manner (Table 3). Here we find that relay A31ACQ24VDC1 with BC148 transistor and flyback diode works out to be the cheapest option that fulfills our requirements.

In the next stage, we look for the analogue voltage sense options. We need to sense three voltages, battery, current in, and current out. We have the option of using a microcontroller based ADC, or we can use an ADC chip. Some microcontrollers have only one or two ADC. We have a third option of using an analogue multiplexer to extend the ADC port of such microcontrollers. Table 4 gives these options and the resulting cost till Stage 3.

Let us look into CPU options in Stage 4. Our expected production volume is low. For such low-volume requirements, it is better to use off-the-shelf CPU modules. We have options of using Arduino Nano, NodeMCU, ESP8266, ESP32, or Raspberry Pi Zero W. Cumulative best cost of these options are listed in Table 5.

Raspberry Pi is a single-board

SD		Options	
Inputs	Arduino Nano	₹557	SD module ₹868
	ESP32	₹569	SD module ₹880
	Node MCU	₹399	SD module ₹710
	R.Pi	₹1,715	In-built ₹1,715
Output		Cloud client	Stand-alone server
		₹710	₹1,715

computer. It comes with its operating system and ports for a keyboard and monitor. We evaluate this option for future requirements.

We look into the option to add Wi-Fi connectivity in the next stage. Arduino needs a Wi-Fi module. Other CPU options have built-in Wi-Fi. Table 6 gives the options and cumulative cost till Stage 5.

Next, we need to add an SD card for data storage. With Raspberry Pi, the SD card is in-built. Other options will need to add an SD card reader module and an SD card. The cumulative cost of these is listed in Table 6.

Conclusion

This analysis helps us to find the most cost-effective design solution for our charge controller. For our product, the most cost-effective configuration works out to be the A31AC-Q24VDC1 relay, BC148 transistor, and flyback diode, NODE MCU module with a CD 4051 analogue multiplexer, and SD card with a reader.

You might have noted that after the analysis we landed up with cost figures that are different from our original assumptions. We need to add other miscellaneous details and re-validate our business case. In addition, we also need to make our design robust. We shall discuss technical risk analysis and tricks to make our product robust in the next article. **EFY**

Anyone who stops learning is old, whether at twenty or eighty.

—Henry Ford

Amaze Yourself!

Read Electronics For You

You'll realise how much you didn't know, and didn't know that you didn't know!

To Subscribe:
<http://subscribe.efy.in>

Scan This Code

